

Compactness Proof for Discrete Navier–Stokes Solutions

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1 Compactness of Discrete Solutions

This section rigorously proves the compactness of discrete solutions $T_h \mathbf{u}_h$ in the transition from discrete to continuous frameworks of the Navier–Stokes equations.

1.1 Theorem Statement

Theorem 1 (Compactness). *Let $\mathbf{u}_h$ be uniformly bounded in $L^2(0, T; H^1(\Omega))$ and assume that the time discretization ensures boundedness of $(\mathbf{u}_h^{n+1} - \mathbf{u}_h^n)/\Delta t$ in $L^2(0, T; H^{-1}(\Omega))$. Then the interpolated solutions $T_h \mathbf{u}_h$ form a compact subset of $L^2(0, T; L^2(\Omega))$.*

1.2 Preliminaries

1.2.1 Sobolev Embedding and Compactness

The Sobolev embedding theorem ensures that for a bounded domain $\Omega \subset \mathbb{R}^3$:

$$H^1(\Omega) \hookrightarrow L^2(\Omega), \tag{1}$$

where $\hookrightarrow$ denotes a compact embedding. This implies that any bounded sequence in $H^1(\Omega)$ has a subsequence that converges strongly in $L^2(\Omega)$.

1.2.2 Aubin–Lions Lemma

The Aubin–Lions lemma provides a framework for compactness in time-dependent function spaces:

Let X, B, Y be Banach spaces such that $X \hookrightarrow B \hookrightarrow Y$. For any sequence $\{v_k\} \subseteq L^2(0, T; X)$ with $\partial_t v_k$ bounded in $L^2(0, T; Y)$, there exists a subsequence that converges strongly in $L^2(0, T; B)$.

In our case:

- $X = H^1(\Omega)$,
- $B = L^2(\Omega)$,
- $Y = H^{-1}(\Omega)$.

1.3 Proof of Compactness

We proceed in four steps, explicitly verifying the conditions of the Aubin–Lions lemma.

1.3.1 Step 1: Boundedness in $L^2(0, T; H^1(\Omega))$

The discrete velocity field $\mathbf{u}_h$ satisfies:

$$\|\mathbf{u}_h\|_{L^2(0, T; H^1(\Omega))} \leq C, \quad (2)$$

where C depends on the initial data and the forcing term $\mathbf{f}$. The interpolation operator T_h preserves these bounds up to discretization error:

$$\|T_h \mathbf{u}_h\|_{L^2(0, T; H^1(\Omega))} \leq C. \quad (3)$$

1.3.2 Step 2: Control of Time Derivatives

The discrete time derivative $(\mathbf{u}_h^{n+1} - \mathbf{u}_h^n)/\Delta t$ is assumed to be bounded in $L^2(0, T; H^{-1}(\Omega))$:

$$\left\| \frac{T_h \mathbf{u}_h(t + \Delta t) - T_h \mathbf{u}_h(t)}{\Delta t} \right\|_{H^{-1}(\Omega)} \leq C. \quad (4)$$

This ensures weak compactness of the temporal changes of $T_h \mathbf{u}_h$.

1.3.3 Step 3: Compact Embedding $H^1(\Omega) \hookrightarrow L^2(\Omega)$

By the Sobolev embedding theorem, $H^1(\Omega)$ is compactly embedded in $L^2(\Omega)$. Therefore, any bounded sequence in $H^1(\Omega)$ has a subsequence that converges strongly in $L^2(\Omega)$. This property guarantees that the spatial components of $T_h \mathbf{u}_h$ satisfy the compactness condition required by the Aubin–Lions lemma.

1.3.4 Step 4: Applying the Aubin–Lions Lemma

Since $T_h \mathbf{u}_h$ is bounded in $L^2(0, T; H^1(\Omega))$ and its discrete time derivative is bounded in $L^2(0, T; H^{-1}(\Omega))$, the Aubin–Lions lemma implies that $T_h \mathbf{u}_h$ has a subsequence that converges strongly in $L^2(0, T; L^2(\Omega))$.

1.4 Remarks and Error Bounds

Remark 1 (Interpolation Errors). *The interpolation operator T_h introduces an error proportional to the grid size h . Formally, for a smooth function $\mathbf{u}$:*

$$\|T_h \mathbf{u}_h - \mathbf{u}\|_{H^1(\Omega)} \leq Ch, \quad (5)$$

where C depends on the smoothness of $\mathbf{u}$. These errors vanish as $h \rightarrow 0$.

Remark 2 (Role of Compactness). *Compactness ensures that the discrete solutions $T_h \mathbf{u}_h$ converge strongly in L^2 , which is essential for handling nonlinear terms like $(\mathbf{u} \cdot \nabla) \mathbf{u}$ in the Navier–Stokes equations.*

1.5 Conclusion

The compactness of $T_h \mathbf{u}_h$ in $L^2(0, T; L^2(\Omega))$ is established. This property plays a crucial role in proving the convergence of discrete solutions to continuous ones and is foundational for establishing global smoothness of the Navier–Stokes equations.